Lab3 Task 1A

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1 Check Even or Odd

This script demonstrates reading user input, performing an arithmetic operation (modulo), and using conditional logic to determine if a number is even or odd.

1.1 Algorithm

1. Start the program.
2. Read the value of n (the number).
3. Calculate the remainder r: $r = n\%2$.
4. If the value of r equals 0 then print the number is even.
5. If the value of r not equal to 0 then print the number is odd.

2 Check Prime Number

This script demonstrates using loops, arithmetic operations, and conditional logic to determine whether a given number is prime. A prime number is a positive integer greater than 1 that has no positive divisors other than 1 and itself.

2.1 Algorithm

1. Start the program.
2. Read the value of n (the number).
3. If $n \leq 1$, then it is not a prime number.
4. Initialize a counter variable `i=2`.
5. Repeat while $i \leq n/2$:
   (a) Calculate: r: $r = n\%2$.
   (b) If $r = 0$, then n is not a prime number; terminate the loop.
   (c) Increment i by 1.
6. If no divisor was found, the number is prime.

3 Find the Factorial of a Number

This script demonstrates using loops and arithmetic operations to compute the factorial of a given number. The factorial of a number n (written as n!) is the product of all positive integers less than or equal to n.

3.1 Algorithm

1. Start the program.
2. Read the value of n.
3. Initialize `fact=1`.
4. Repeat while $n > 0$:
   (a) Multiply `fact = fact * n`.
   (b) Decrement `n = n - 1`.
5. Print the value of `fact`.

4 Display Fibonacci Series

This script demonstrates using loops, variables, and arithmetic operations to generate a sequence of Fibonacci numbers. The Fibonacci sequence starts with 0 and 1, and each subsequent number is the sum of the previous two.

4.1 Algorithm

1. Start the program.
2. Read the number of terms, n.
3. Initialize `a=0, b=1`.
4. Print `a` and `b`.
5. Repeat $(n - 2)$ times:
   (a) `c = a + b`.
   (b) Print `c`.
   (c) Update: `a = b`, `b = c`.

5 Find the Largest of Three Numbers

This script demonstrates the use of conditional statements (`if`, `elif`, `else`) to find the largest among three given numbers.

5.1 Algorithm

1. Start the program.
2. Read three numbers: `a, b, c`.
3. Compare:
   (a) If `a > b` and `a > c`, then `a` is the largest.
   (b) Else if `b > c`, then `b` is the largest.
   (c) Else `c` is the largest.
4. Display the largest number.

6 Check Whether a Year is Leap Year or Not

This script demonstrates conditional logic and modulus arithmetic to check leap year rules.

6.1 Algorithm

1. Start the program.
2. Read the year value `y`.
3. If `y % 400 == 0`, then it is a leap year.
4. Else if `y % 100 == 0`, then it is not a leap year.
5. Else if `y % 4 == 0`, then it is a leap year.
6. Else, it is not a leap year.
7. Display the result.